

Erdős Problem #917

EP-917

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 4$ and $f_k(n)$ be the largest number of edges in a graph on n vertices which has chromatic number k and is critical (i.e. deleting any edge reduces the chromatic number).

Is it true that $f_k(n) \gg_k n^2$? Is it true that $f_6(n) \sim n^2/4$? More generally, is it true that, for $k \geq 6$, $f_k(n) \sim \frac{1}{2} \left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>